# Reproducing DoRA for Commonsense Reasoning with Student-Scale Compute

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## Motivation & Paper Idea

Full fine-tuning large language models is expensive in memory, storage, and training time. **LoRA** reduces this cost by training low-rank adapter updates while freezing the base model, but the low-rank update alone may limit how pretrained weights can change.[1]

### DoRA

- **DoRA** decomposes pretrained weights into **magnitude** and **direction**.[3]
- Direction receives a **LoRA**-style low-rank update; magnitude is learned separately.
- The paper reports improved accuracy over **LoRA** across multiple tasks.

### Project goal

- Central question: does **DoRA** improve commonsense reasoning accuracy over a same-scope **LoRA** baseline?
- Reproduce the paper's LLaMA2-7B commonsense reasoning trend.
- Test whether **DoRA** helps most in attention layers, MLP layers, or only when applied broadly.

## Methodology

### Model, data, and metrics

- Base model: `meta-llama/Llama-2-7b-hf`.
- Training data: `commonsense_15k.json`, a student-scale commonsense subset.
- Evaluation: BoolQ, PIQA, Social IQA, HellaSwag, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA.
- Metrics: per-task accuracy and macro-average accuracy across all eight tasks.

### Implementation

- From-scratch PyTorch **LoRA** and **DoRA** adapter layers injected into Hugging Face model modules.
- Frozen base model, adapter-only checkpoints, 3 epochs, cutoff length 256.
- 4-bit NF4 Colab runtime presets for T4, L4, and A100 GPUs.

**Controlled comparison.** Same model, 15k data subset, tasks, prompt/eval pipeline, seed 42, and adapter-only training. Paper-style method settings: **LoRA** r=32, LR=3e-4; rank-halved **DoRA** r=16, LR=2e-4.

## Our Independent Experiment: Scope Ablation

| Scope | Target modules | Purpose |
|---|---|---|
| **Full** | `q,k,v,up,down` projections | Paper-level reproduction |
| **Attention only** | `q,k,v` projections | Test routing/context layers |
| **MLP only** | `up,down` projections | Test feed-forward layers |

**Motivation.** This ablation tests whether **DoRA**'s gain comes from attention layers, MLP layers, or coordinated updates across both.

**Design.** Each scope compares same-scope **LoRA** and **DoRA** adapters under the same evaluation pipeline.

## Headline Result

| 77.41% | 79.20% | +1.79 | 6/1/1 |
|---|---|---|---|
| Full **LoRA** macro | Full rank-halved **DoRA** macro | accuracy points | task wins/ties/losses |

Figures 1 and 2 show that full-scope, rank-halved **DoRA** reproduced the paper's qualitative improvement over **LoRA** under our student-scale setup.

## Macro Accuracy by Adapter Scope

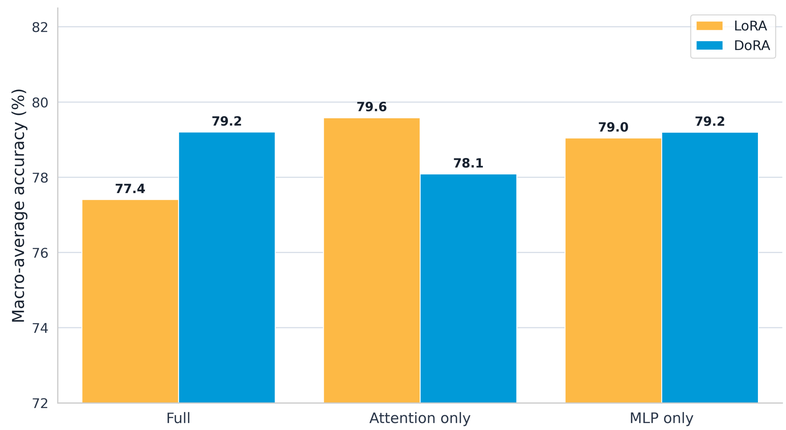


**Figure 1.** Full-scope, rank-halved **DoRA** improves macro accuracy from 77.4% to 79.2%. Attention-only **DoRA** does not improve over its same-scope **LoRA** baseline.

| Scope | LoRA macro | DoRA macro | Delta |
|---|---|---|---|
| Full | 77.41% | **79.20%** | **+1.79** |
| Attention only | **79.59%** | 78.09% | -1.50 |
| MLP only | 79.05% | **79.20%** | +0.15 |

## Where Does DoRA Help?

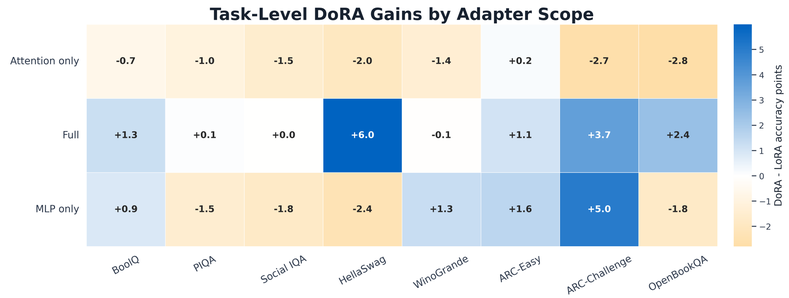


**Figure 2.** Task-level **DoRA** minus **LoRA** gains. Full-scope **DoRA** improves 6/8 tasks, ties Social IQA, and slightly loses on WinoGrande.

**Largest full-scope gains.** HellaSwag +6.0, ARC-Challenge +3.7, OpenBookQA +2.4 accuracy points.

**Takeaway.** **DoRA**'s benefit appears scope-dependent, not automatic.

## Official Reference Context

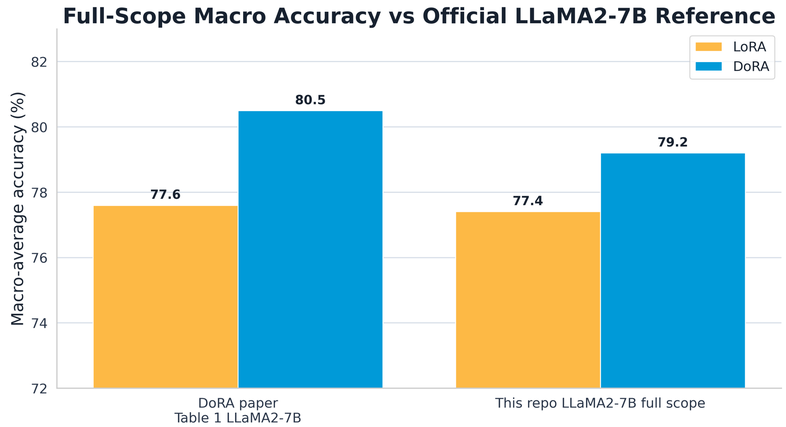


**Figure 3.** Our result matches the paper's direction: rank-halved **DoRA** improves over **LoRA**, but our 79.2% **DoRA** score remains below the official 80.5% reference.

## Interpretation

- The main paper-level claim **partially reproduced**: full-scope **DoRA** beat full-scope **LoRA**.
- Scope ablations show the gain was **not automatic**: attention-only **DoRA** underperformed **LoRA** on 7/8 tasks; MLP-only **DoRA** was roughly tied.
- Key takeaway: **DoRA** appears most useful when attention and MLP transformations adapt together, not as a replacement for one module group.
- Key limitations: 15k subset instead of full data, seed 42 only, quantized Colab runs, and a local implementation rather than the exact official recipe.

## Conclusion & Future Work

### Takeaways

- Full-scope **DoRA** improved macro-average accuracy by **+1.79 points**.
- The reproduction supports the qualitative value of weight-decomposed low-rank adaptation.
- The ablation adds a caution: applying **DoRA** narrowly can remove the benefit.

### Next steps

- Train on the full `commonsense_170k.json` dataset.
- Repeat all six conditions across multiple random seeds.
- Add parameter-count, GPU-memory, and wall-clock cost comparisons.

## References

[1] Edward J. Hu et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[2] Hugging Face. Transformers documentation. `https://huggingface.co/docs/transformers`, 2026.

[3] Shih-Yang Liu et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. Oral presentation.

[4] NVIDIA Research. NVlabs/DoRA: Official implementation. `https://github.com/NVlabs/DoRA`, 2024.